

Graph Representation Learning

In this project, you are expected to investigate the properties of graph neural networks with a specific focus on their properties, including *inductive bias*, *information propagation*, *expressive power*, and *generalization*. The task is to develop and study a well-defined research problem based on certain desiderata of your choice pertaining to one (or more) of these properties and report your theoretical and/or empirical findings in a controlled experimental setup. The study can focus on one (or more) of these dimensions of your choice, including (but not limited to):

- Analyzing different choices of model components and their implications on the property of interest (e.g., *how does a choice for a model component affect expressive power?*),
- Exploring model design space and deriving architectures in an application domain (e.g., *how does a choice for a model affect its inductive bias on molecular graphs?*),
- Comparing two (or more) graph architectures with respect to a property, and analyzing their strengths or weaknesses (e.g., *how does one model generalize compared to another model?*),
- Comparing properties of graph neural networks to other architectures (e.g., Tabular models, LLMs, etc.) in a specific setting (e.g., *how does the inductive bias of tabular foundation model compare to a graph foundation model on simple graphs? What are the (dis)advantages?*),
- Proposing an extension (or simplification) of an existing graph neural network architecture, and studying its properties (e.g., *how does the information flow in a newly proposed architecture and what are the (dis)advantages?*),
- Analyzing the properties of variations of graph neural networks on different types of input graphs (e.g., multi-relational, planar, directed graphs, temporal, random graphs) in a specific setting (e.g., *do graph transformers with positional encodings allow for better information flow and inductive bias on simple graphs?*),
- Identifying a limitation of existing architectures and quantifying this limitation both empirically and theoretically (e.g., *what are the limitations of knowledge graph foundation models in terms of their expressive power?*).

Report Please use the official L^AT_EX template (available in TISS) to write your report. The report should not exceed *five* pages using the margins and font size of the provided L^AT_EX template, *excluding* the page(s) for the references, and supplementary data (e.g., implementation details). The report should include a link to your anonymous code-base (i.e., a github repository¹). The structure of the report will have the form of a small workshop paper (see, e.g., the provided L^AT_EX template as a reference). For example, the report can start with an introduction motivating the problem (1. Introduction). This can be followed with a section where the core methodology and the overall approach is reported (2. Methods). The main findings can be reported in a separate section (3. Results). The report can then conclude with a discussion on the limitations and future directions (4. Conclusions). You can introduce further sections or subsections as you see necessary or even modify existing sections.

¹You can anonymize your github repository at: <https://anonymous.4open.science>.

As part of this report, you are expected to:

- (a) State a *small* research question related to the properties of graph neural networks and provide a clear motivation for the proposed study. (*Example: How does model architecture A compare to model architecture B in terms of expressive power? Why is it important to answer this question?*)
- (b) Propose means to study the research question and give an outline of the overall approach, clearly identifying the goals of the chosen study. (*Example: An experimental setup which aims to test a hypothesis through, e.g., a synthetic dataset or using existing datasets, as well as a theoretical justification for the study.*)
- (c) Clearly state your methodology, empirical setup, benchmarks, hyper-parameters, and the assumptions underpinning the study.
- (d) Report your empirical/theoretical/conceptual findings, identifying whether or not the results support the initial hypothesis. You should use visuals, figures, and tables to present your findings in a structured manner.
- (e) Relate and compare the obtained results to the relevant literature and to the concepts from the course.
- (f) Present a detailed discussion relating the results to the original motivation, as well as a critical perspective regarding the study.
- (g) Provide an outlook stating any additional studies that need to be conducted to reach to more conclusive statements.

Assessment criteria The study will be assessed based on the following criteria: (i) motivation, clarity, and presentation (15 marks), (ii) coherence and depth of the study (15 marks), (iii) originality and novelty (e.g., in the problem setup, or in the approach, or in the results) (10 marks), (iv) the presence of a balanced critical self-evaluation (8 marks), (v) relations to the concepts discussed throughout the course (6 marks), (vi) scholarship, i.e., its framing in the existing literature (6 marks). Please ensure that the computational resources are used in a balanced and well-justified manner: *The study is not evaluated according to the number of experiments, benchmarks, or datasets, but according to its merits in accordance with the outlined criteria.* Strong submissions typically focus on one clearly defined property or hypothesis in a carefully scoped setting. Studies with well-motivated hypotheses and carefully conducted analyses will be rewarded even when the results are negative.

(60 marks)